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PROGRAMMING WITH PYTHON

Lekha A

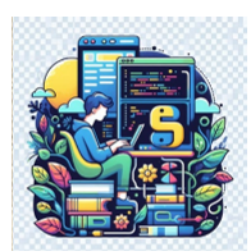
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PROGRAMMING WITH PYTHON

Python Basics and Standard Library Functions

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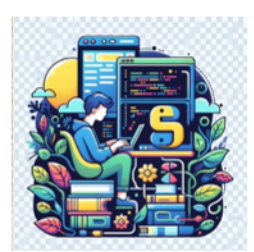
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Functions

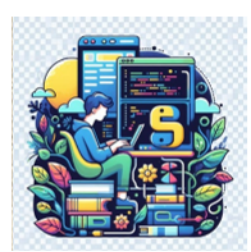
- A function is a block of organized, reusable code that is used to perform a single, related action.
- It is a named section of a program that performs a specific task.
- Functions provide better modularity for the application and a high degree of code reusing.



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Functions

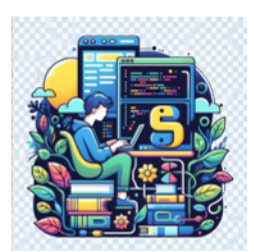
- Function blocks begin with the keyword **def** followed by the function name and parentheses ().
 - The first statement of a function can be an optional statement - the documentation string of the function or docstring.
 - The code block within every function starts with a colon (:) and is indented.



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Functions

- Any input parameters or arguments should be placed within these parentheses.
- The statement `return [expression]` exits a function, optionally passing back an expression to the caller.
 - A return statement with no arguments is the same as `return None`.

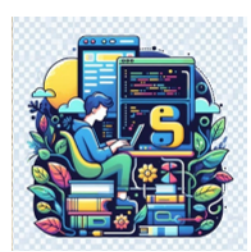


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Functions

- By default, parameters have a positional behavior and have to be informed in the same order that they were defined.

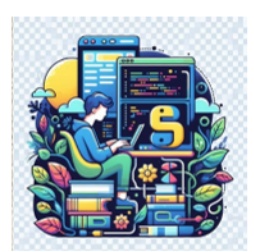
- `def functionname( parameters ):`
- `"function_docstring"`
- `function_suite`
- `return [expression]`



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Calling Functions

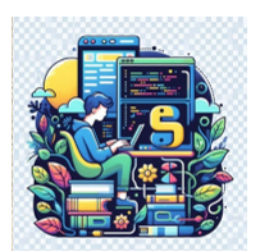
- Defining a function only gives it a name, specifies the parameters that are to be included in the function and structures the blocks of code.
- Once the basic structure of a function is finalized, execute it by calling it from another function or directly from the Python prompt.



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Pass by reference vs. Pass by value

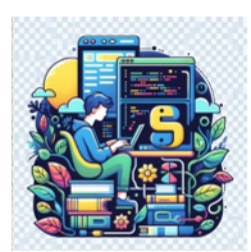
- Pass by reference
 - It is used in some programming languages, where address of the variable is passed and then the operation is done on the value stored at these addresses.



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Pass by reference vs. Pass by value

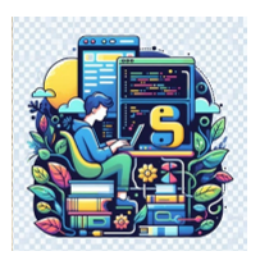
- Pass by value
 - It means that the value is directly passed as the value to the argument of the function.
 - Here, the operation is done on the value and then the value is stored at the address.
 - Pass by value is used for a copy of the variable.



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Pass by reference vs. Pass by value

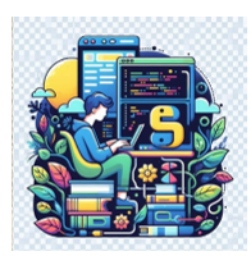
- All parameters (arguments) in the Python language are passed by reference.
- If what a parameter refers to within a function is changed, the change also reflects back in the calling function.
- The parameter has no specified type.
- The type is decided when the argument is copied on call to the function at runtime – type is determined dynamically.



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Arguments

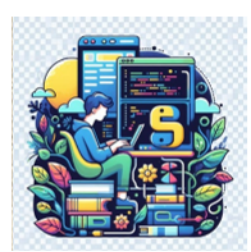
- Required arguments
- Keyword arguments
- Default arguments
- Variable-length arguments



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Required Arguments

- Required arguments are the arguments passed to a function in correct positional order.
- The number of arguments in the function call should match exactly with the function definition



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Required Arguments

```
def printme( name, age ):  
    "This prints a passed string into this function"  
    print (name , age)
```

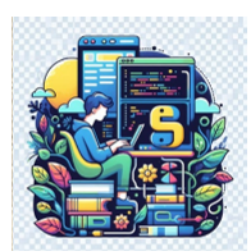
```
printme("Ajay",30)
```

Ajay 30

```
printme("Ajay")
```

```
-----  
TypeError                        Traceback (most recent call last)  
Cell In[3], line 1  
----> 1 printme("Ajay")
```

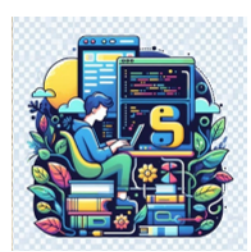
TypeError: printme() missing 1 required positional argument: 'age'



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Keyword Arguments

- Keyword arguments are related to the function calls.
- When keyword arguments are used in a function call, the caller identifies the arguments by the parameter name.



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Keyword Arguments

```
def printme( name, age ):  
    "This prints a passed string into this function"  
    print (name , age)
```

```
printme(age=30, name="Ajay")
```

Ajay 30

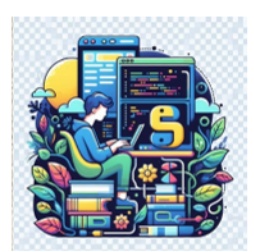
```
printme(name="Ajay")
```

TypeError Traceback (most recent call last)

Cell In[9], line 1

```
----> 1 printme(name="Ajay")
```

TypeError: printme() missing 1 required positional argument: 'age'



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Default Arguments

- A default argument is an argument that assumes a default value if a value is not provided in the function call for that argument.

```
def printme( name, age=40 ):
    "This prints a passed string into this function"
    print (name , age)
```

```
printme("Ajay")
```

Ajay 40

```
printme("Ajay", 30)
```

Ajay 30



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Variable-Length Arguments

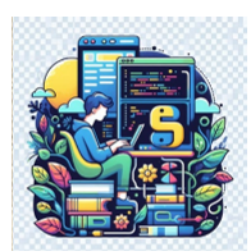
- A function may process more arguments than specified number of arguments.
- These arguments are called variable-length arguments and are not named in the function definition,
- An asterisk (*) is placed before the variable name that will hold the values of all non-keyword variable arguments.



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Variable-Length Arguments

- This tuple remains empty if no additional arguments are specified during the function call.
- `def functionname ([formal_args,] *var_args_tuple ):`
- `"function_docstring"`
- `function_suite`
- `return [expression]`



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Variable-Length Arguments

```
def multiplier(*num):  
    prod = 1  
    for i in num:  
        prod = prod * i  
    print("Product:", prod)
```

```
multiplier(3,5)
```

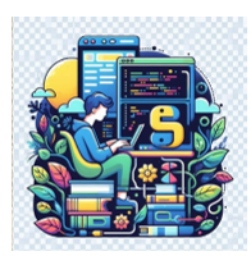
Product: 15

```
multiplier(1,2,4)
```

Product: 8

```
multiplier(2,2,6,7)
```

Product: 168



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Recap

- Functions
- Arguments to a function



THANK YOU

Lekha A
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